Kick-off Diagnostic · L2-6 — The Quadratic Formula

Grade 10

8 questions · 30 seconds each · about 4 minutes · Diagnose phase (6 min of 60)

Pre-session anchor (slides 2 and 3 of the pre-session must say this, so items 2 and 6 test it).

The one idea: Write a, b, c with their signs, then $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$; the discriminant $b^2 - 4ac$ under the root tells you how many real solutions there are.

The mistake most students make: Keeping b instead of $-b$ (writing -7 when $b = -7$), or squaring $(-7)^2$ as -49.

Readiness items 3, 4, 5, 7, 8 test: solving by factoring and square roots, completing the square (Lesson 2-5), vertex form and graph reading, and the sign of the discriminant from a parabola's x-intercepts.

Wayground set-up: timer 30 s on every question; shuffle options on, questions off; hide the leaderboard; show answers after the quiz, not after each item. Item 1 is a poll (no points). Items 2 to 8 are 1 point each. Tags below: **maroon** = tells you about the pre-session, **navy** = readiness for today. **Green** = correct.

1 POLL · 30 s — SELF-REPORT

Be honest: how much of the pre-session task did you do?

A. All four slides, including the readiness check

B. I watched and read but skipped the check

C. I opened it but did not finish

D. I did not open it

Signal. Ungraded. Tally it first; it is the self-report half of the pre-session test.

2 MULTIPLE CHOICE · 30 s — PRE-SESSION CHECK

In the pre-session, the solutions of $ax^2 + bx + c = 0$ are…

A. $x = \dfrac{b \pm \sqrt{b^2 - 4ac}}{2a}$

✓ **B.** $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

C. $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2}$

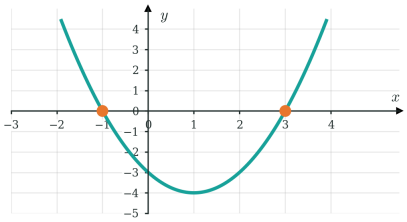
D. $x = -b \pm \dfrac{\sqrt{b^2 - 4ac}}{2a}$

Signal. Wrong → skipped slide 2. A = lost the minus on b; C = divided by 2, not $2a$; D = only the root over $2a$.

3 FILL IN THE BLANK · 30 s — READINESS

The graph of $y = x^2 - 2x - 3$ is shown. The solutions of $x^2 - 2x - 3 = 0$ are $x = -1$ and $x =$ _____

Accept: 3



Signal. -3 = read the y-intercept; 1 = axis of symmetry; -4 = lowest y-value. Reading zeros is not yet secure.

4 CHECKBOX · 30 s — READINESS

Select the **two** correct statements.

A. $x^2 = 49$ gives only $x = 7$

✓ **B.** $(x - 1)^2 = 25$ **gives** $x = 6, -4$

✓ **C.** $x^2 - 5x + 6 = 0$ **gives** $x = 2, 3$

D. $x^2 + 3x = 0$ gives only $x = -3$

Signal. Choosing A → forgot ± when taking a square root; D → divided by x and lost the solution $x = 0$.

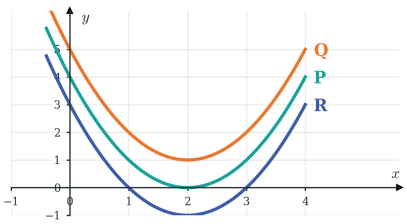
5 MATCH · 30 s — READINESS

Curves P, Q and R are $y = x^2 - 4x + c$ for three values of c. Match each curve with its number of real solutions of $x^2 - 4x + c = 0$.

Curve P → **one**

Curve Q → **none** (shown shuffled in Wayground)

Curve R → **two**



Signal. P touches the axis. Mixing P and Q = treating a touch as no solution.

6 MULTIPLE CHOICE · 30 s **PRE-SESSION CHECK**

For $2x^2 - 7x + 3 = 0$, which answer is the mistake the pre-session warned about?

A. $x = \dfrac{7 \pm 5}{2}$

B. $x = \dfrac{7 \pm \sqrt{73}}{4}$

C. $x = \dfrac{7 \pm 5}{4}$

✓ **D.** $x = \dfrac{-7 \pm 5}{4}$

Signal. Slide 3's "mistake most students make" (D). A = divided by 2, not $2a$; B = $49 + 24$; C is the correct answer.

7 REORDER · 30 s **READINESS**

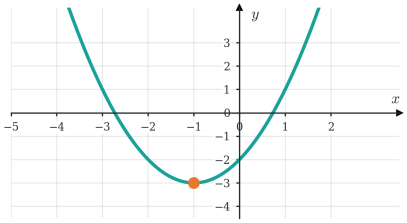
Put the steps for solving $x^2 + 4x - 1 = 0$ by completing the square in order.

Correct order: $x^2 + 4x = 1 \rightarrow x^2 + 4x + 4 = 5 \rightarrow (x + 2)^2 = 5 \rightarrow x = -2 \pm \sqrt{5}$

Signal. Wrong → completing the square (Lesson 2-5) is the gap; the formula comes from exactly these steps.

8 DROPDOWN · 30 s **READINESS**

The graph is a parabola with its vertex marked. Which is its vertex form?



✓ **A.** $y = (x + 1)^2 - 3$

B. $y = (x - 1)^2 - 3$

C. $y = (x + 1)^2 + 3$

D. $y = (x - 3)^2 + 1$

Signal. Read the vertex $(-1, -3)$. B = sign of h reversed; C = sign of k reversed; D = coordinates swapped.

Read the results (90 seconds)

Readiness R — **Items 3, 4, 5, 7, 8 (out of 5). 0–2** → re-teach negative substitution, radicals and completing the square with me. **3–4** → straight to the formula and the discriminant; fix the missed item in passing. **5** → start the Investigate prompt now.

Pre-session P — **Items 2 and 6 (out of 2), read with the item 1 poll.**

What you see	Reading	What you do
Poll "all" or "read", $P = 2$	Prepared	Nothing extra. Use their R score to route.
Poll "all" or "read", $P = 0$	Claimed, did not retain	Two-minute recap of the one idea at the start. Ask them to restate it in their own words.
Poll "opened" or "did not open"	Task not done	No penalty. Pair with a prepared partner for Targeted Instruction; note it for your task tracker.
$R \leq 2$ and $P = 2$	Content gap	Re-teach group. The task was done, so the idea needs a different explanation.
$R \geq 4$ and $P \leq 1$	Task skipped, skills fine	Move on. Chase the task, not the content.

Class-level rule. If more than 30% of the class chose "opened" or "did not open", or the class average on P is below 1, spend three minutes recapping the one idea above before Targeted Instruction. If more than 30% score $R \leq 2$, the whole class takes the re-teach route first.